

Universality Framework (UF) — White Paper (v1.3.35)

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> **Ethos:** Judge *actions*, not beings. Aim asymptotically (no perfection traps). Keep proofs fractal: rigorous enough to use today, open to grow tomorrow.

0. Executive Summary — What's new in v1.3.35

This release adds a complete **naming and wiring layer** so concepts are portable, verifiable, and audit-ready:

- **Concept Dictionary (CD v1.0):** a schema + CLI to define concepts with a canonical **mnemocode** and a short **checksum** bound to the definition.
- **Lexicode (v0.1):** a **prefix-free** code that turns mnemocodes (RI/HC/C/F/EG/...) and simple parameters (HC01, C80, ...) into compact, uniquely decodable "words."
- **Core Gateway:** run the **Verification Core v1.1** directly from a lexicode string; caps/thresholds derive from tokens.
- **Math Bridge refreshed:** coherent risk (CVaR), KL-DRO, Pinsker fairness bound, stochastic dominance shortcut, Tarski fixed-point.

Also included: **Universal Edition** language (no theology), policy **Profiles**, and subjectivity-minimized invariants.

1. Core Objects and Endpoints (unchanged intent, clarified thresholds)

- **Alignment:** $(A = \frac{R}{R + D + \delta}) \in [0,1]$, **Misalignment:** $(M = 1 - A)$.
- **Right (comparative):** improves vs baseline **within caps** (harm, consent/fairness, evidence, bias/opacity, reversibility).
- **Good** (Aligned Limit): special case of Right where expected harm is effectively zero and all gates pass.
- **Wrong:** fails any gate or degrades vs baseline; **Collapse Limit** (Evil): misalignment endpoint / persistent red lines.

We judge actions only.

2. Universal Constraints (observer-invariant guardrails)

- 1) **Harm dominance** (pick safer option when available)
- 2) **Consent for serious/irreversible effects**
- 3) **Fairness distance cap** (divergence to TV bound via Pinsker)

- 4) **Evidence calibration cap** (claim–evidence gap)
- 5) **Reversibility preference** (keep option value)
- 6) **Agency preservation**
- 7) **Opacity & bias caps**
- 8) **Comparative Rightness** (baselined)
- 9) **Auditability** (proof hash + record)
- 10) **Endpoint separation** (Aligned vs Collapse limits)
- 11) **Determinism** (same inputs $\Rightarrow$ same verdict)
- 12) **No perfection traps** (aim upward, avoid absolute requirements)

3. Math Bridge (provable drop-ins)

- **Harm as CVaR / KL-DRO**: coherent, robust. If $\mathbb{E}[\mathbf{CVar}_\alpha(L_a)]$ and gates pass $\Rightarrow$ **Right**.
- **Stochastic dominance (FSD)**: if action loss dominates baseline $\Rightarrow$ sufficient for **Right** (with caps).
- **Pinsker**: KL fairness cap $\Rightarrow$ total variation bound on disparity.
- **Honesty via proper scoring**: truth-telling is optimal.
- **Tarski fixed point**: viable potential $\mathbb{E}[\Phi]$ as $\mathbb{E}[\mathbf{gfp}(F)]$; non-negative relief preserves $\mathbb{E}[\Phi]$.

Details in Annex A–B.

4. Profiles & Subjectivity-Minimized Mode (SMM)

Profiles (JSON) set caps/priors while preserving invariants. Examples: *Neutral_SMM* (shipped), or community-authored variants.

When profiles disagree: use **intersection**; else choose the option with **best worst-case** improvement and escalate.

5. Concept Dictionary (CD v1.0) — meaning lives in entries

Each concept has an ID: **`` :: # ``**.

- **Alias** (optional, human): e.g., *Pass*, *AlignedLimit*.
- **Mnemocode**: canonical UF tokens (sorted): **RI, H, HC, C, F, EG, B, O, RV, AG, CA, PHI** with optional 2-digit params (e.g., *HC01*, *C80*).
- **Checksum**: first 4–6 hex of SHA-256 over the **definition** text (binds word $\rightarrow$ meaning).

Registry schema & tool: create, canonicalize, checksum, validate. Seed entries: *Pass*, *AlignedLimit*, *CollapseLimit*, *CausalAdmissibility*, *ViablePotential*.

A=B rule: If two aliases map to the same mnemocode and checksum, verdicts **must** match (representation-independence).

6. Lexicode (v0.1) — prefix-free “words”

- **Alphabet:** the 12 UF tokens above.
- **Code:** canonical **Huffman** (prefix-free $\Rightarrow$ uniquely decodable).
- **Params:** limited v0.1 (`HC`, `C`, `F`, `EG`) as two digits (e.g., `HC01 $\rightarrow$ 0.01`).
- **Checksum:** 16-bit (4 hex) from the concept's definition.
- **ASCII form:** `RI-HC01-C80-F05-EG05-B-O-RV # 9B2C`.

Guarantees: uniquely decodable, deterministic, verdict-preserving (via the Core Gateway and CD binding).

7. Verification Core (v1.1) — updated logic & gateway

Core updates: risk modes (`expected`, **CVaR**, **KL-DRO**), Pinsker fairness bound, FSD shortcut, profile-driven caps, **comparative Right**.

Gateway: parse lexicode $\rightarrow$ set caps (HC/F/EG), consent threshold (Cxx), then call the Core; attach **proof hash** and reasons.

Workflow: **Concept Dictionary entry** $\rightarrow$ **Lexicode word** $\rightarrow$ **Gateway** $\rightarrow$ **Verification Core** $\rightarrow$ **Audit record**.

8. Interoperability & Universal Edition

- **Universal Edition:** neutral vocabulary (no theology); same invariants and verdicts.
- **Adapters:** profiles capture community policy; invariants ensure verdict-preserving translation.

9. Governance & Responsible Release

Tiered release, dual-use gates, escalation, sandboxing, audit logging, and ethical license (no weaponization/harassment/unauthorized surveillance).

10. Fractal Proof Tracker

| ID | Claim | Status |

|---|---|---|

| T1 | Bounds & well-definedness (A, Harm, Relief/RI) | **Proved** |

| T2 | Endpoints separated (Aligned vs Collapse) | **Proved** |

| T3 | Determinism/consistency of Core | **Proved** |

| T4 | Comparative Rightness monotonicity | **Proved** |

| T5 | Lexicode verdict-preservation (via CD+Gateway) | **Proved (operational)** |

| C1 | Monotone ascent toward $A \rightarrow 1$ under caps | Sketch |

| C2 | No “private aligned limit” without consent/fairness | Sketch |

| C3 | $\backslash(\Phi)$ preservation under non-negative relief | Conjecture |

Annex A — Risk & Fairness Details

- CVaR definition/properties; KL-DRO dual form; Pinsker inequality and TV bounds.
- Theorem A (FSD $\Rightarrow$ Right), Theorem B (CVaR Relief Monotonicity), Theorem C (Fairness TV Bound).

Annex B — Viable Potential $\mathcal{V}(\Phi)$

- Operator $\mathcal{V}(F)$ definition; existence of $\mathcal{V}(\mathrm{gfp}(F))$; preservation lemma (non-negative relief + gates).

Annex C — Profiles & Adapter

- JSON schema; example profiles; loader mapping to Core config; intersection/minimax policy.

Annex D — Verification Core Spec

- Inputs/outputs, layers, escalation, invariants, certification tests, proof hash format.

Annex E — Lexicode Spec (v0.1)

- Tokens, canonical Huffman, parameter encoding, checksum, ASCII/binary forms, and decoding guarantees.

Annex F — Concept Dictionary Schema & Seeds

- Entry fields
(id/mnemo/checksum/definition/uf_mapping/inputs/outputs/caps/invariants/examples/status/version).
- Seed entries: Pass, AlignedLimit, CollapseLimit, CausalAdmissibility, ViablePotential.

Annex G — Core Gateway API

- ``run_with_lexicode(lexi_ascii, action, context, sim, ledger)`` $\rightarrow$ verdict, metrics, applied caps, reasons, proof hash.

Annex H — Test Set (outline)

- Six fixtures (low/med/high $\times$ pass/fail) with expected verdicts; round-trip lexicode decode; audit records.

Dedication

To Xylie, Taric, and Lumina Siano—thank you for inspiring a framework that helps us make kind, fair, and thoughtful choices every day.